

D1. A simple batch distillation separates 0.6 kmol of a binary feed that is 70.0 mol% methanol and 30.0 mol% water. The final still pot is 10.0 mol% methanol. Find W_{fin} , distillate D_{tot} and $x_{D,\text{avg}}$. VLE data are in Table 2-8. $p = 1.0$ atm.

a. Use Simpson's rule.

b. Use the Gaussian quadrature formula.

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$F = 0.6 \text{ kmol}$ $x_F = 0.7$ $x_f = 0.1$ $x_{\text{mid}} = \frac{0.7+0.1}{2} = 0.4$

x	y	$\frac{1}{y-x}$
0.7	0.81	5.88
0.4	0.729	3.04
0.1	0.418	3.14

a) $\text{area} = \frac{0.7-0.1}{6} [3.14 + 4(5.88) + 5.88]$

$\text{area} = 2.118$

$W_f = F e^{-\text{area}} = 0.6 e^{-2.118} = 0.072 \text{ kmol}$

$D_{\text{tot}} = F - W_f = 0.6 - 0.072 = 0.528 \text{ kmol}$

$x_{D,\text{avg}} = \frac{F x_F - W_f x_f}{D_{\text{tot}}} = \frac{(0.6)(0.7) - (0.072)(0.1)}{0.528} = 0.78$

b) $h = (x_F - x_f) / 2 = (0.7 - 0.1) / 2 = 0.3$ $x_{\text{mid}} = 0.4$

$\text{area} = \frac{h}{9} [5 \cdot f(x_{\text{mid}} - h\sqrt{0.6}) + 8 \cdot f(x_{\text{mid}}) + 5 \cdot f(x_{\text{mid}} + h\sqrt{0.6})]$

$x_{\text{mid}} - h\sqrt{0.6} = 0.4 - 0.2324 = 0.1676$

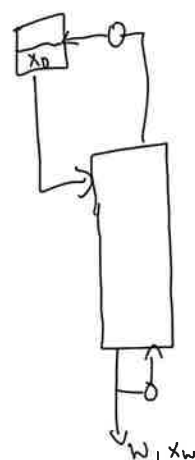
$x_{\text{mid}} + h\sqrt{0.6} = 0.4 + 0.2324 = 0.6324$

x	y	$\frac{1}{y-x}$
0.1676	$0.517 + \left(\frac{0.1676-0.15}{0.05}\right)(0.519-0.517) = 0.5581$	2.694
0.4	0.729	3.04
0.6324	$0.825 + \left(\frac{0.6324-0.6}{0.05}\right)(0.8475-0.825) = 0.8346$	4.826

$\text{area} = \frac{0.3}{9} [5(2.694) + 8(3.04) + 5(4.826)] = 2.064$ $W_f = 0.6 e^{-2.064} = 0.076 \text{ kmol}$

$D_{\text{tot}} = 0.6 - 0.076 = 0.524 \text{ kmol}$ $x_{D,\text{avg}} = \frac{(0.6)(0.7) - (0.076)(0.1)}{0.524} = 0.79$

D3.* A distillation system with a still pot plus a column with one equilibrium stage is used to batch distill 1.0 kmol of a 57.0 mol% methanol and 43.0 mol% water feed. A total condenser is used. The final bottoms concentration is 15.0 mol% methanol. Pressure is 101.3 kPa. Reflux is a saturated liquid, and $L/D = 1.85$. Find W_{final} , D_{total} and $x_{D,\text{avg}}$. VLE data are in Table 2-8.



2 stages

$$\frac{L}{D} = 1.85$$

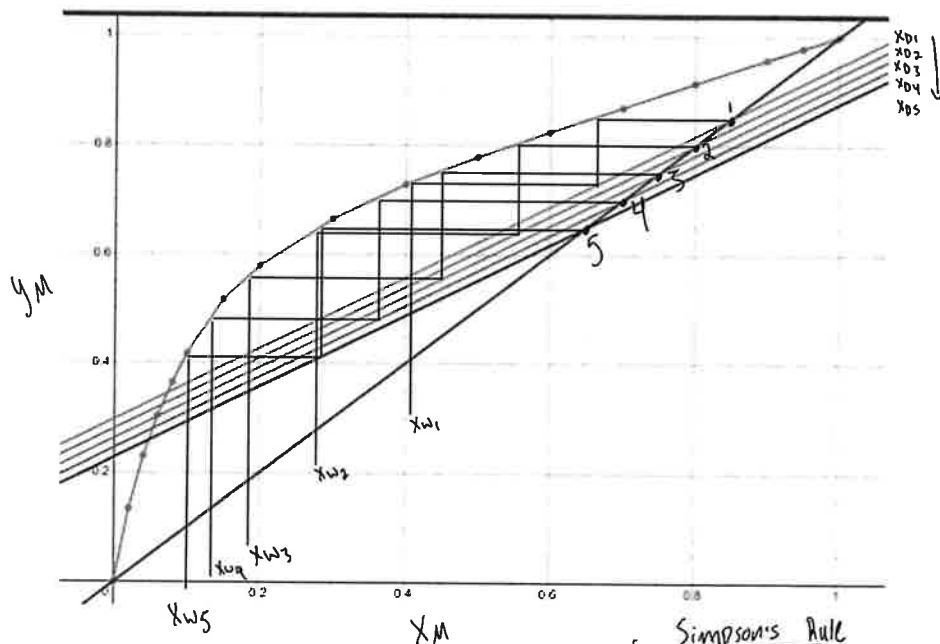
$$x_F = 0.57$$

$$x_B = 0.15$$

op eqn: $y = \frac{L}{V}x + (1 - \frac{L}{V})x_D$

slope: $\frac{L}{V} = \frac{L}{L+D} = \frac{L/D}{L/D+1} = \frac{1.85}{1.85+1} = 0.65$

y intercept: $(1 - \frac{L}{V}) = 1 - 0.65 = 0.35 \rightarrow y = 0.65x + 0.35x_D$



	x_D	x_W	$\frac{1}{x_D - x_W}$
1:	0.85	0.41	2.273
2:	0.80	0.275	1.905
3:	0.75	0.18	1.754
4:	0.70	0.14	1.786
5:	0.65	0.10	1.818

1: $y = 0.65x + 0.2975$

2: $y = 0.65x + 0.28$

3: $y = 0.65x + 0.2625$

4: $y = 0.65x + 0.245$

5: $y = 0.65x + 0.2275$

$x_{\text{mid}} = \frac{x_F + x_B}{2} = \frac{0.57 + 0.15}{2} = 0.36$ Simpson's Rule

$$\int_{x_F}^{x_B} g(x) dx = \frac{(x_F - x_B)}{6} [g(x_B) + 4g(x_{\text{mid}}) + g(x_F)]$$

$$= \frac{0.57 - 0.15}{6} [1.905 + 4(2.089) + 2.273] = 0.926$$

$$W_F = F e^{-0.926} = 39.6 \text{ kmol}$$

$$F = W_F + D_{\text{tot}} \rightarrow D_{\text{tot}} = F - W_F = 100 - 39.6 = 60.4 \text{ kmol}$$

$$x_{D,\text{avg}} = \frac{F \cdot x_F - W_F \cdot x_B}{D_{\text{tot}}} = \frac{(100)(0.57) - (39.6)(0.15)}{60.4} = 0.845$$

2. A simple batch still is separating a feed that is 60.0 mol% 1,2 dichloroethane and 40.0 mol% 1,1,2-trichloroethane. Pressure is 1 atm. The relative volatility is constant, $\alpha_{1,2} \approx 2.4$. The use of Eq. (9-13) is recommended.

- The charge (F) is 1.3 kmol. The final still pot concentration is 30.0 mol% 1,2 dichloroethane. Find the final moles in the still pot and the average mole fraction of distillate product.
- Repeat part a if $F = 3.5$ kmol.
- If $F = 2.0$ kmol and the average distillate concentration is 75.0 mol% 1,2-dichloroethane, find final kmols in the still pot and the final mole fraction of 1,2 dichloroethane in the still pot.

Eqn (9-13): $\ln\left(\frac{W_F}{F}\right) = \frac{1}{\alpha-1} \ln\left(\frac{x_F(1-x_F)}{x_F(1-x_F)}\right) + \ln\left(\frac{1-x_F}{1-x_F}\right)$ [when α constant]

a) $F = 1.3$, $x_F = 0.6$, $\alpha_{D-T} = 2.4$, $x_F = 0.3$

$$\ln\left(\frac{W_F}{1.3}\right) = \frac{1}{2.4-1} \ln\left(\frac{0.3(1-0.6)}{0.6(1-0.3)}\right) + \ln\left(\frac{1-0.6}{1-0.3}\right) = -0.895 + -0.5596 = -1.454$$

$$\ln\left(\frac{W_F}{1.3}\right) = -1.454 \rightarrow \frac{W_F}{1.3} = 0.234 \rightarrow W_F = 0.3037 \text{ kmol}$$

$$D_{\text{tot}} = 1.3 - 0.3037 = 0.9963 \text{ kmol} \quad x_{D,\text{avg}} = \frac{F \cdot x_F - W_F \cdot x_F}{D_{\text{tot}}} = \frac{(1.3)(0.6) - (0.3037)(0.3)}{0.9963} = 0.6914$$

b) $\ln\left(\frac{W_F}{3.5}\right) = -1.454 \rightarrow \frac{W_F}{3.5} = 0.234 \rightarrow W_F = 0.819 \text{ kmol}$

$$D_{\text{tot}} = F - W_F = 3.5 - 0.819 = 2.681 \text{ kmol} \quad x_{D,\text{avg}} = \frac{(3.5)(0.6) - (0.819)(0.3)}{2.681} = 0.6916$$

c) $F = 2$, $x_F = 0.6$, $x_{D,\text{avg}} = 0.75$

$$D_{\text{tot}} \cdot x_{D,\text{avg}} = F \cdot x_F - W_F \cdot x_F \rightarrow (F - W_F) x_{D,\text{avg}} = F \cdot x_F - W_F \cdot x_F \rightarrow F x_{D,\text{avg}} - W_F x_{D,\text{avg}} = F \cdot x_F - W_F \cdot x_F$$

$$F x_{D,\text{avg}} - F x_F = W_F x_{D,\text{avg}} - W_F x_F \rightarrow F(x_{D,\text{avg}} - x_F) = W_F(x_{D,\text{avg}} - x_F)$$

$$\frac{W_F}{F} = \frac{x_{D,\text{avg}} - x_F}{x_{D,\text{avg}} - x_F} \rightarrow \ln\left(\frac{x_{D,\text{avg}} - x_F}{x_{D,\text{avg}} - x_F}\right) = \frac{1}{\alpha-1} \ln\left[\frac{x_F(1-x_F)}{x_F(1-x_F)}\right] + \ln\left(\frac{1-x_F}{1-x_F}\right)$$

$$\ln\left(\frac{0.75-0.6}{0.75-x_F}\right) = \frac{1}{1.4} \ln\left[\frac{0.4x_F}{0.6(1-x_F)}\right] + \ln\left[\frac{0.4}{1-x_F}\right] \rightarrow e^{\ln\left(\frac{0.15}{0.75-x_F}\right)} = e^{\ln\left[\left(\frac{0.4x_F}{0.6(1-x_F)}\right)^{1/1.4} \cdot \frac{0.4}{1-x_F}\right]}$$

1st guess $x_F = 0.3$: $0.3333 = 0.3075$

2nd guess $x_F = 0.4$: $0.4286 = 0.3736$

3rd guess $x_F = 0.5$: $0.6 = 0.5988$ ✓

$$\frac{0.15}{0.75-x_F} = \left(\frac{0.4x_F}{0.6-0.6x_F}\right)^{1/1.4} \cdot \frac{0.4}{1-x_F}$$

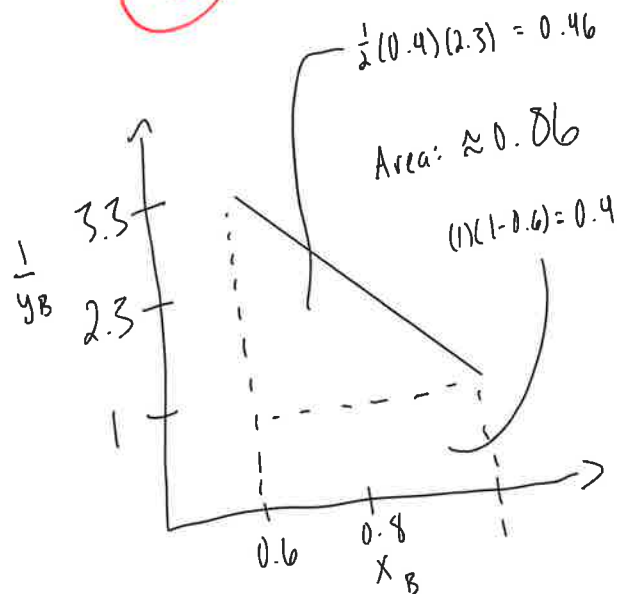
$$W_F = F \cdot \frac{x_{D,\text{avg}} - x_F}{x_{D,\text{avg}} - x_F} = 2 \cdot \frac{0.75-0.6}{0.75-0.5} = 1.2 \text{ kmol} \quad x_F = 0.5$$

D11. A constant-mole batch distillation is used to change from a pure n-butanol solvent to a solvent that is 60.0 mol% n-butanol and 40.0 mol% water. If the initial charge is $W = 2.0$ kmol, how much water, S , must be added to achieve the desired solvent concentration? Do the integration with Simpson's rule, dividing the entire integration from 0.6 to 1.0 first into one step and then into two steps. Compare the answers. 2

$$X_{B+OH,i} = 1 \quad X_{B+OH,f} = 0.6$$

* referred to table 8-5

X_B	X_W	y_B	$\frac{1}{y_B}$
1	0	1	1
0.8	0.2	0.435	2.3
0.6	0.4	0.3	3.33



$$\text{area} = \frac{1-0.6}{6} [3.33 + 4(2.3) + 1] = 0.902$$

*not accurate enough

$$\frac{S}{W} = 0.92 \rightarrow S = 2 \cdot 0.902 = 1.804 \text{ kmol}$$

X_B	X_W	y_B	$\frac{1}{y_B}$
0.9	0.1	0.58	1.724
0.7	0.3	0.34	2.941

$$\text{area} \int_{0.8}^1 = \frac{1-0.8}{6} [2.3 + 4(1.724) + 1] = 0.34$$

$$\text{area} \int_{0.6}^{0.8} = \frac{0.8-0.6}{6} [3.33 + 4(2.941) + 2.3] = 0.58$$

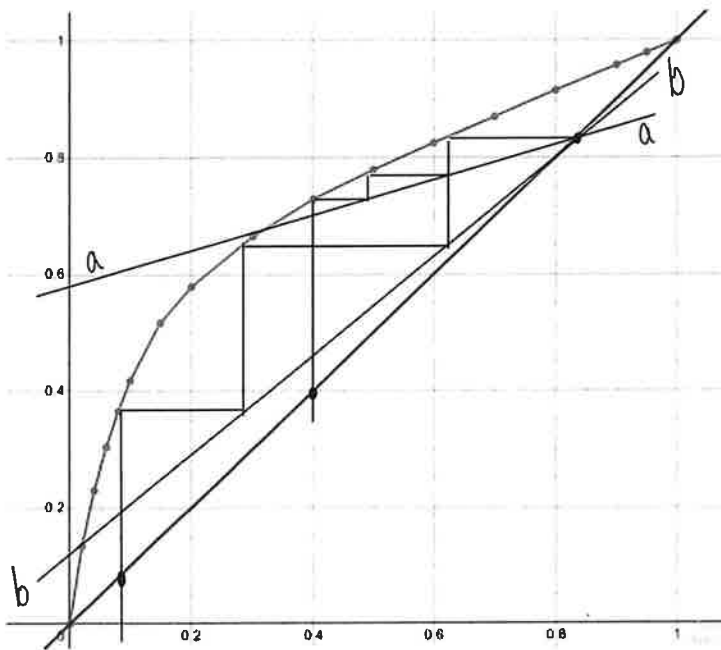
$$\text{total area: } 0.92$$

$$S = 2 \cdot 0.92 = 1.84 \text{ kmol}$$

D16.* 10.0 kmol of a 40.0 mol% methanol and 60.0 mol% water mixture is processed in a normal batch distillation system with a still pot that acts as an equilibrium stage and a column with two equilibrium stages (total of three equilibrium contacts). The column has a total condenser, and reflux is a saturated liquid. The column is operated with a varying reflux ratio so that x_D is held constant. We desire a final still pot concentration of 8.0 mol% methanol, and distillate should be 84.0 mol% methanol. Pressure is 1 atm, and CMO is valid.

- What initial external reflux ratio, L_0/D , must be used?
- What final external reflux ratio must be used?
- How much distillate product is withdrawn, and what is the value of W_{final} ?

$F = 10 \text{ kmol} \quad x_F = 0.4 \quad x_f = 0.08 \quad x_D = 0.84$



a) $x_F = 0.4$ to $x_D = 0.84$

$$\frac{L}{V} = \frac{0.84 - 0.57}{0.84 - 0} = 0.32$$

$$\frac{L}{D} = \frac{L}{V-L} = \frac{L/V}{1 - L/V} = \frac{0.32}{1 - 0.32} = 0.47$$

b) $x_f = 0.08$ to $x_D = 0.84$

$$\frac{L}{V} = \frac{0.84 - 0.13}{0.84} = 0.845$$

$$\frac{L}{D} = \frac{0.845}{1 - 0.845} = 5.45$$

c) $F = D_{\text{tot}} + W_f \rightarrow 10 = D_{\text{tot}} + W_f$

$F x_F = D_{\text{tot}} \cdot x_D + W_f \cdot x_f \rightarrow (10)(0.4) = 0.84D + 0.08W_f$

$$\begin{aligned} 4 &= 0.84D + 0.08W_f \\ W_f &= 10 - D \end{aligned} \quad \left. \begin{aligned} 4 &= 0.84D + 0.08(10 - D) \rightarrow 4 = 0.84D + 0.8 - 0.08D \\ 3.2 &= 0.76D \rightarrow D = 4.21 \text{ kmol} \end{aligned} \right\}$$

$W_f = 10 - 4.21 = 5.79 \text{ kmol}$

* a E b
done by
trial & error

(2)